Applied Surface Science 753 (2027) 168134

Contents lists available at ScienceDirect

Applied Surface Science

journal homepage: www.elsevier.com/locate/apsusc

Full Length Article

Surface modification of cellulose nanofibril films via organosilicon deposition using dielectric barrier discharge plasma (DBD) plasma for packaging applications

Mirela A. Artner [a], Lucia Švandová [b,c], Nathalie Lavoine [a,*], Jacopo Profili [d,*]

[a] Department of Forest Biomaterials, North Carolina State University, 2820 Faucette Drive, Raleigh, NC 27695, USA
[b] Department of Plasma Physics and Technology, Masaryk University, Kotlarska 267/2, 611 37 Brno, Czech Republic
[c] R&D Center for Plasma and Nanotechnology Surface Modifications, CEPLANT, Kotlarska 267/2, 611 37 Brno, Czech Republic
[d] Department of Biology, Chemistry and Geography, University of Quebec in Rimouski, Quebec, Canada

ARTICLE INFO

Keywords:
Cellulose nanofibrils
Dielectric barrier discharge
Plasma
Barrier properties
Organosilicon coating

ABSTRACT

Films made of cellulose nanofibrils (CNFs) are promising sustainable alternatives to petroleum-based plastic packaging due to their gas and grease barrier properties, high strength-to-flexibility ratio, tunable transparency, and biodegradability. However, cellulose's hygroscopic nature limits its direct use in food packaging. Here, we exploit dielectric barrier discharge (DBD) plasma to tune the surface properties of CNF films and improve their water/moisture barrier performance. Two strategies were tested: (i) static-mode deposition of organosilicon fragments using 2,4,6,8-tetramethylcyclotetrasiloxane (TMCTS) as precursor, and (ii) formation of multilayers with varied surface chemistries by adjusting the plasma carrier gas and organosilicon fragment chemistry. Strategy (i) identified process parameters enabling uniform deposition of hydrophobic TMCTS fragments on CNF film surfaces. Strategy (ii) extended this by creating a multilayer structure combining tailored inorganic (hydrophilic) and organic (hydrophobic) layers, which substantially reduced water absorption and improved protection relative to untreated films. These results demonstrate that optimizing plasma gas composition, treatment time, and application mode (static vs. dynamic) allows controlled deposition of chemical moieties, enabling CNF films with tunable water and water–vapor barrier properties suited to diverse food-packaging requirements. Plasma treatment thus offers a sustainable, scalable route to enhance CNF film performance and expand their use in packaging applications.

1. Introduction

Global and concerted efforts towards the development of sustainable alternatives to petroleum-based plastics have been driven by the increasing demand for low-cost, environmentally safe, and non-toxic packaging materials. Out of the ~6 million tons of plastic packaging being produced in 2021, 36 % were labelled as reusable, recyclable or compostable. Yet only 13.3 % out of these 2.2 million of packaging labelled as "sustainable" ended up being recycled [1]. Multilayer packaging such as flexible packaging is the most impacted category by these end-of-life challenges. Multilayer packaging substrates are manufactured through the smart association of different plastic films (*e.g.,* polyethylene (PE)) and non-plastic materials (*e.g.,* aluminum foil) to create unique properties that cannot be achieved by a single material [2]. For instance, PE is commonly utilized for its moisture and gas barrier capabilities, while aluminum is privileged for its protection against light, ultraviolet radiation, and overall ability to extend the shelf-life of the food product. Although the manufacturing process of multilayer packaging offers immense benefits with respect to weight-to-performance and performance-to-cost ratios, we are not yet equipped to separate these thin layers of materials for the proper recycling of flexible packaging [3]. One solution to address this challenge lies in the replacement of the non-recyclable/non-biodegradable layers with sustainable alternatives without compromising the overall package's functionality.

Cellulose nanofibrils (CNFs) represent a unique solution for multilayer packaging design and manufacturing [4]. Films made of CNFs have shown promise as sustainable packaging because they come from

* Corresponding authors.
E-mail addresses: nmlavoin@ncsu.edu (N. Lavoine), profili.jacopo@gmail.com (J. Profili).

https://doi.org/10.1016/j.apsusc.2026.168134
Received 8 July 2025; Received in revised form 22 July 2026; Accepted 15 August 2026
Available online 8 September 2026
0169-4332/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

renewable resources, exhibit excellent mechanical properties [5], and high barrier properties against oxygen and grease [6,7]. Despite these advantages, the use of CNF films in the packaging industry is slowed down because of their inherent sensitivity to moisture and water. Specifically, non-modified CNF films exhibit a much higher water vapor transmission rate (WVTR), commonly exceeding 200 g/m^2.day at 50% relative humidity (RH) [4] than the commercial plastic substrates they aim to replace, such as PE films with a WVTR of 16.5 g/m^2.day at 100% RH [8]. Several strategies have been explored to improve the WVTR of CNF films and involve, among others, the grafting of aromatic polymers like polystyrene [9], esterification of cellulose using anhydrides of different carbon chain lengths [10], acetylation of cellulose by *e.g.,* N,N-dimethylacetamide [11], and silylation [12]. These strategies have been quite effective in lowering the water vapor permeability and water uptake of CNF films. For instance, acetylation of CNFs resulted in an 80% decrease of the films' water uptake and reduced the films' WVTR at 50% RH from 234 g/m^2.day (unmodified films) to 167 g/m^2.day (acetylated films) [11]. Such considerable improvements yet remained insufficient to compete with commercial plastic films [13]. Other promising strategies that resulted in even lower WVTR values [14,15], while promising, usually involve the use of toxic chemicals, solvents, or high temperatures, or require multiple reaction steps. Certain chemical reactions were also reported to alter, and potentially impair the inherent properties of cellulose such as crystallinity and strength [16,17], and its biodegradability [18]. Hence, there is a need for more sustainable and efficient approaches to tailor the surface chemistry of CNF films without altering their bulk properties.

The use of sustainable processing for the functionalization of materials' surface has hence become more attractive for addressing water and moisture sensitivity of lignocellulosic biomass, as these so-called green technologies are seen as more eco-friendly and, in some cases, more time and energy efficient than chemical modification approaches. Dielectric barrier discharge (DBD) plasma can be used to modify the surface properties of lignocellulosic substrates without compromising their bulk performance. Functionalization by plasma does not necessarily require the use of toxic chemicals, solvents, or water [19] and can be conducted under atmospheric pressure and at low temperature conditions [20]. In addition, plasma is a highly versatile and scalable technique that can be used for finetuning the surface chemistry of substrates by proper selection of the plasma gas and adapted to roll-to-roll manufacturing of *e.g.,* flexible packaging [21,22]. Previous research has reported successful use of DBD plasma to tailor the wettability of cellulosic surfaces. Coplanar diffusive DBD plasma combined with atmospheric air and nitrogen gases was used to increase the moisture absorption and wettability of newspapers for improved printability [23]. The treatment successfully led to a decrease in the water contact angle (WCA) from 134° to 49° after plasma treatment, indicative of significant surface oxidation. In contrast, the treatment of chlorine-free bleached softwood pulp with DBD air plasma was found to decrease the wettability of the pulp by enhanced surface roughness [24].

In addition to varying the gas composition and plasma geometry to generate different types of surface functionalization, plasma can be used for the deposition of thin molecular layers. The deposition of plasma-induced organosilicon fragments onto the surface of lignocellulosic substrates has especially received considerable attention because of its versatility in surface energy modification, durability, stability in different environments, and ability to endow substrates with effective surface barrier properties [25]. WCA values higher than 120° were achieved after DBD plasma treatment (atmospheric pressure, 20 kHz, static condition) of a Kraft paper using 1,3,5,7-tetramethylcyclotetrasiloxane (TMCTS) as the organosilicon precursor and helium as the carrier gas [21]. The resulted ~1-μm-thick TMCTS coating, however, exhibited uncoated or partially coated regions onto the paper surface, highlighting the strong influence of the substrate's morphology (*i.e.,* the presence of entangled, layered, macrofibrils) on the uniformity, morphology and efficiency of the plasma-deposited coating. The uncoated and partially coated areas reduced the water barrier performance of the treated paper substrates due to partial delamination caused by rapid water uptake at these sites and possible reduced adhesion between the plasma-treated layers. One approach to enhance adhesion between layers of organosilicon consists in tuning the surface chemistry of the coating by altering the gas composition that reacts with the organosilicon precursor [26]. The incorporation of oxidizing gases such as oxygen (O_2), air, and nitrous oxide (N_2O) in the plasma gas line can eliminate the hydrophobic, organic portion (*e.g.,* methyl groups) of the organosilicon coating to enhance the hydrophilicity of the treated surface, such as polycarbonate [22]. Significant progress has been made on plasma-induced deposition of organosilicon fragments onto cellulosic substrates [21,27–29], but there is still a need for deciphering the plasma-induced deposition mechanism onto CNF films [30,31]. Because CNF films usually exhibit a lower porosity, a more entangled structure, and a higher surface area with more accessible functional groups than their macroscale counterparts [17,32], their interaction with plasma may differ from what has been observed with wood fibers, textiles, and paper substrates.

In this study, we present a scalable and sustainable all-plasma strategy to tailor the surface properties of CNF films via DBD, aiming to enhance their performance in moisture-sensitive packaging. While plasma hydrophobization is known, the novelty of this work lies in the controlled architectural design of single- and multi-layer coatings relying exclusively on gas-phase plasma processing. Specifically, this work introduces the comparison between static and dynamic deposition modes to decouple the spatial fragmentation pathways of the TMCTS precursor and their subsequent impact on film morphology and chemistry. In addition, a continuous, dry-coating in-situ strategy is developed by alternating hydrophobic and oxidized hydrophilic layers, significantly improving interfacial adhesion and coating integrity without wet chemistry. Finally, a comprehensive mapping is established linking plasma fragment distribution to surface wettability and macro-scale moisture barrier performance. This approach offers a highly tunable, energy-efficient pathway to functionalize nanocellulose substrates via dry processes compatible with industrial roll-to-roll manufacturing.

2. Experimental section

2.1. Materials

Never-dried Northern Bleached Softwood Kraft (NBSK) pulp was provided by WestRock (Covington, VA, USA). The pulp was rinsed and washed with deionized water, then centrifuged and fluffed to remove the excess of water and for long-term storage in refrigerated conditions. 2,4,6,8-tetramethylcyclotetrasiloxane (TMCTS, $O_4Si_4(CH_3)_4H_4$, NMR grade, purity ≥99.5 vol%) and anhydrous calcium chloride ($CaCl_2$) (powder, ≥97%) were all purchased from Sigma-Aldrich and used as received.

2.2. Production of cellulose nanofibrils (CNFs) and CNF films

A 10 wt.% NBSK pulp was refined using a PFI mill refiner (the Norwegian Pulp and Paper Institute, Norway) at 5000 rpm according the T248 sp-00, (2000) standard [33]. The refined pulp was passed through a Masuko supermasscolloider grinder (MKCA6-5J®, Sangyo, Japan) at a consistency of 1.7 wt.% using a rotation speed of 1200 rpm with varying gap clearance and number of passes, to reach a total energy consumption of 3.61 kWh/kg of oven-dry pulp. The produced suspension was then further processed through an aqueous counter collision (ACC Starburst, Sugino Machine Co., Ltd, Japan) at a jet pressure of 220 MPa, for three passes, to produce an aqueous suspension of CNFs at 0.1 wt.%. The resulting CNFs had an average width of 15–30 nm, determined by Transmission Electron Microscopy (**Fig. S1**).

Self-standing films of CNFs were produced by water evaporation of a 0.1 wt.% aqueous CNF suspension poured into polystyrene Petri dishes

of 8.5 mm in diameter. An average film grammage of 10 g/m^2 was targeted. After complete drying at ambient conditions, the films were conditioned at 23 °C and 50% relative humidity (RH) for at least 72 h prior to testing.

2.3. Dielectric barrier discharge (DBD) plasma treatment

The DBD reactor (**Fig. S2**) was built with two parallel plate electrodes: the upper plate was a 0.64-mm-thick alumina sheet coated with conductive paint (3.5 × 3.0 cm^2), and the bottom one was a stainless-steel plate (13 × 9 cm^2) (Figs. 1 **and S2**). Plasma was generated by the application of a sinusoidal voltage with a peak-to-peak amplitude of 18 kV and a frequency of 3 kHz. The discharge was applied with a 1-mm gap between the electrodes and the CNF film. The system operated with two gas lines: one for nitrogen, N_2 (carrier gas at 3 L/min), and another for nitrous oxide, N_2O, (2 mL/min). The flow rates were monitored using a Bronkhorst™ controller (USA).

The CNF films were treated by plasma in static and dynamic modes using 2,4,6,8-tetramethylcyclotetrasiloxane (TMCTS) as the siloxane precursor. In static mode, the bottom plate and CNF films to be treated were kept fixed (Fig. 1). The plasma treatment time was set at 10 min. 15 ppm of TMCTS was suspended in nitrogen gas and introduced in the reactor. In dynamic mode, the bottom plate was moved back and forth at a speed of 35 cm/min for continuous deposition of TMCTS using N_2 gas, for a total amount of time of 30 min (1st layer). A second treatment with N_2O (oxidizing gas) and N_2 gas was used to produce a 2nd layer. This second treatment was carried out for one (1) min without the presence of TMCTS. A 3rd layer was finally conducted atop the 2nd layer using the same treatment conditions as the 1st layer (TMCTS precursor, N_2 gas and 30 min treatment time). The CNF films were treated in triplicates for each treatment.

2.4. Characterization of the TMCTS-plasma-induced coating

The plasma-treated CNF films and respective untreated controls were analyzed by Field-Emission Scanning Electron Microscopy (FE-SEM) using a FEI Quanta 250 microscope (Thermo-Fisher Scientific, USA). The acceleration voltage was 12.5 kV, and the working distance was 10 mm. The images were acquired at magnifications ranging from ×500 to ×1000.

Prior to FE-SEM analysis, the films were coated for 1 min with a 10-nm-thick gold/palladium (Au/Pd) alloy layer. A minimum of 5 images were acquired per specimen, and the best acquisitions were used for discussion.

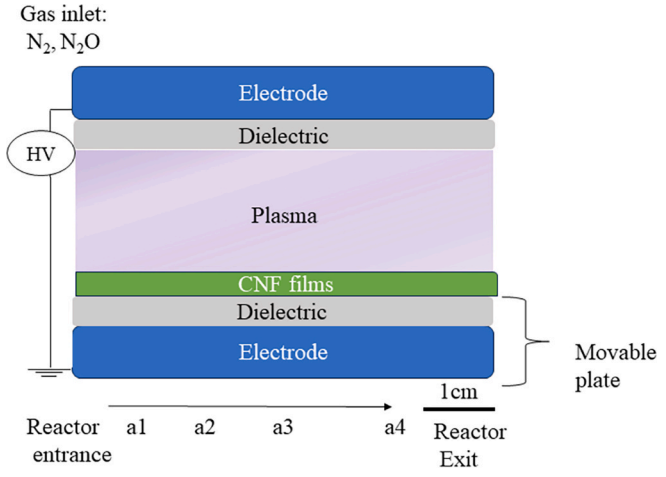


Fig. 1. Schematic of the side view of the DBD plasma reactor. The inlet gases used as precursor carriers were injected on the upper left, at the entrance of the reactor. The different sample areas of deposition were labelled a0, a1, a2, a3, and a4, corresponding to 1, ~1.2, 1.5, and 3 cm away from the reactor entrance, respectively.

2.5. Attenuated total reflectance Fourier transformed infrared spectroscopy (ATR-FTIR)

The surface chemistry of the plasma-treated films and controls was investigated using an FTIR spectrophotometer (Cary 660 FTIR, Agilent Technologies, Victoria) in ATR mode. Infrared spectra were recorded at room temperature at a resolution of 4 cm^{-1} using a 45° incident ATR accessory (1.5 mm^2 active sample area, Silicium crystal with a depth of penetration of 0.170 µm at 3500 cm^{-1}). A total of 128 scans were conducted. For each specimen, 15 equally spaced points were analyzed between the entrance and the exit side of the plasma gas (*i.e.*, between a1 and a4, Fig. 1).

Measurements were made in triplicates using the OriginPro®software (OriginLab® Corporation, Software version 9.9). The spectra were normalized to the IR signal of the Si-O-Si asymmetric stretching band at 1200–1000 cm^{-1}.

2.6. Water Contact Angle (WCA)

The sessile drop method was applied to measure the water contact angle of the plasma-treated films and controls using a Video Contact Angle System (VCA-2500 XETM, AST products Inc., Billerica, MA, USA) equipped with a high-resolution CCD camera. The volume of the ultra-pure water drops was kept at 1 µL.

Measurements were made at least in triplicates for each plasma treatment and the average water contact angle values were used for discussion.

2.7. X-ray Photoelectron Spectroscopy (XPS)

The surface composition of the plasma-treated CNF films and controls was analyzed using an X-ray Photoelectron Spectroscopy (PHI 5600-ci spectrometer, Physical Electronics, Chanhassen, MN, USA) operated under high vacuum (<10^{-6} Pa). A standard Al X-ray source (Kα, hν=1486.6 eV) at 300 W with a charge neutralizer was used to record survey spectra (0–1400 eV). Standard Mg X-ray source (Kα, hν=1253.6 eV) at 300W was used to collect high-resolution spectra (HR XPS). Photoelectron detection was carried out at a take-off angle of 45° for a detection area of 0.5 mm^2.

Surface charging effects were compensated for by correcting the C-C/C-H aliphatic carbon binding energy peak at 285.0 eV. Curve fitting of the C 1s and Si 2p was performed using the Gaussian-Lorentzian functions after Shirley-type background subtraction (CASAXPS Version 2.3.25) (as shown **Fig. S3**). Nine replicates per plasma treatment were analyzed.

2.8. Atomic Force Microscopy (AFM)

AFM analysis was performed using an Asylum research atomic force microscope (Oxford instrument, USA) in tapping mode with an etched silicon tip (OTESPA, tip radius <10 nm). Surface topography was evaluated in triplicates for areas covering 50 × 50 µm^2. The AFM images were analyzed using the Gwyddion v2.65 software and the surface roughness of the films was determined using the Roughness Average (R_a).

R_a is calculated considering the deviation of all points roughness profile from a mean line over the evaluation length of the areas according to ISO 4287/1-1997 standard.

2.9. Water vapor transmission rate (WVTR) and water uptake of CNF films

The WVTR of CNF films was measured at 23 °C and 50% RH ac-

cording to the ASTM E96 (2000) standard [34]. Aluminum cups (68-3000 EZ-Cup, Vapometer, USA) were filled with 5 g of anhydrous calcium chloride ($CaCl_2$). CNF films of 8 cm in diameter (for a tested surface area of 6.25×10^{-4} m^2) were tightly sealed between the cup base and its lid. The increase of the film's mass over time was measured at regular time intervals using an analytical laboratory scale (± 0.0001 g) (Radwag AS-120.R2-PLUS, USA) until a constant change in mass was reached. The experiments were conducted in duplicates for each surface treatment condition, including control films. The WVTR was calculated as follows:

$$WVTR\left(\frac{g}{m^2 day}\right) = \frac{final\ mass\ (g) - initial\ mass\ (g)}{surface\ area\ of\ the\ film\ (m^2)\ *\ total\ number\ of\ days}$$

The water uptake of the films was measured using an adapted Cobb method (TAPPI 441 om-09 (2013) [35]). A metal ring with a test area of 10 cm^2 was used for the measurements. The initial mass of the film was measured using an analytical laboratory scale (± 0.0001 g) (Radwag AS-120.R2-PLUS, USA). The film was tightly clamped between the metal ring and a solid, metallic surface to enable the pouring of 10 mL of distilled water for full contact with the 10 cm^2 area, without leaking. A contact time of 20 s was recorded, after which the film was pressed between two blotting papers and weighed again using the same balance. The tests were performed in triplicates and the average water uptake values are used for discussion. The water uptake was calculated as follows:

$$Water\ Uptake\ (\%) = \frac{final\ mass\ (g) - initial\ mass\ (g)}{initial\ mass\ (g)}\ x\ 100$$

3. Results and discussion

3.1. Influence of organosilicon plasma-induced deposition on CNF films' topography

The CNF films were first functionalized by plasma in static mode (*i.e.*, 10-min treatment per position, with fixed plates) to assess the growth dynamics of organosilicon deposition between the entrance and the exit of the plasma reactor. Fig. 2 shows the scanning electron microscopy (SEM) images of the films' surfaces before and after organosilicon deposition using TMCTS as the organosilicon precursor and nitrogen plasma (TMCTS/N_2). Four different areas of deposition located between the entrance and the exit of the reactor (Fig. 1) were analyzed, namely at 1 cm (a1, Fig. 2**B** and **C**), 1.2 cm (a2, Fig. 2**D**), 1.5 cm (a3, Fig. 2**E**) and 3 cm (a4, Fig. 2**F**) away from the entrance of the reactor. As the CNF films were functionalized further away from the introduction of the carrier gas and TMCTS precursor, their surface morphology changed significantly.

Compared to the well-reported, tight entangled structure of cast, neat CNF films (Fig. 2**A**) [36], the films that were treated closer to the entrance of the reactor (a1 and a2) demonstrated a rougher surface topography that was entirely covered by cauliflower-like spheres (Fig. 2**B**, **C**, and **D**). Such observation was accentuated in the presence of non-fully processed fibers, whose morphology was revealed with the localized, preferred assembly of TMCTS spheres along the fibers' length, as shown in Fig. 2**C**. The observed increase in surface roughness was additionally confirmed by Atomic Force Microscopy (AFM) with the measurement of a roughness average (R_a) of 58 ± 8 nm for the control CNF film against 171 ± 8 nm for the plasma-treated films at the position a1 (Table 1). This phenomenon correlates with the growth mechanism of the coating, primarily driven by the localized nucleation of the reactive species on the regions exposed to higher fragment concentrations (*i.e.*, closer to the middle of the electrode gap) or having specific features (*e.g.*, fiber edges) that potentially enhance the electric field [21].

As we moved towards the exit of the reactor (a3 and a4 positions, Fig. 2**D** and **E**), the surface morphology of the treated CNF films became less and less granular, yet remained rougher than the surface of the non-treated CNF films. These SEM observations were supported by the measurement of the films' roughness: the films in a4 position showed a

Table 1
Average roughness (R_a) values of CNF films before and after plasma treatment at the position 1 cm (a1), 1.5 cm (a3) and 3 cm (a4) away from the reactor entrance. The average roughness values were measured by Atomic Force Microscopy (AFM).

	Untreated CNF films	a1	a3	a4
R_a (nm)	56 ± 9	171 ± 16	78 ± 4	76 ± 8

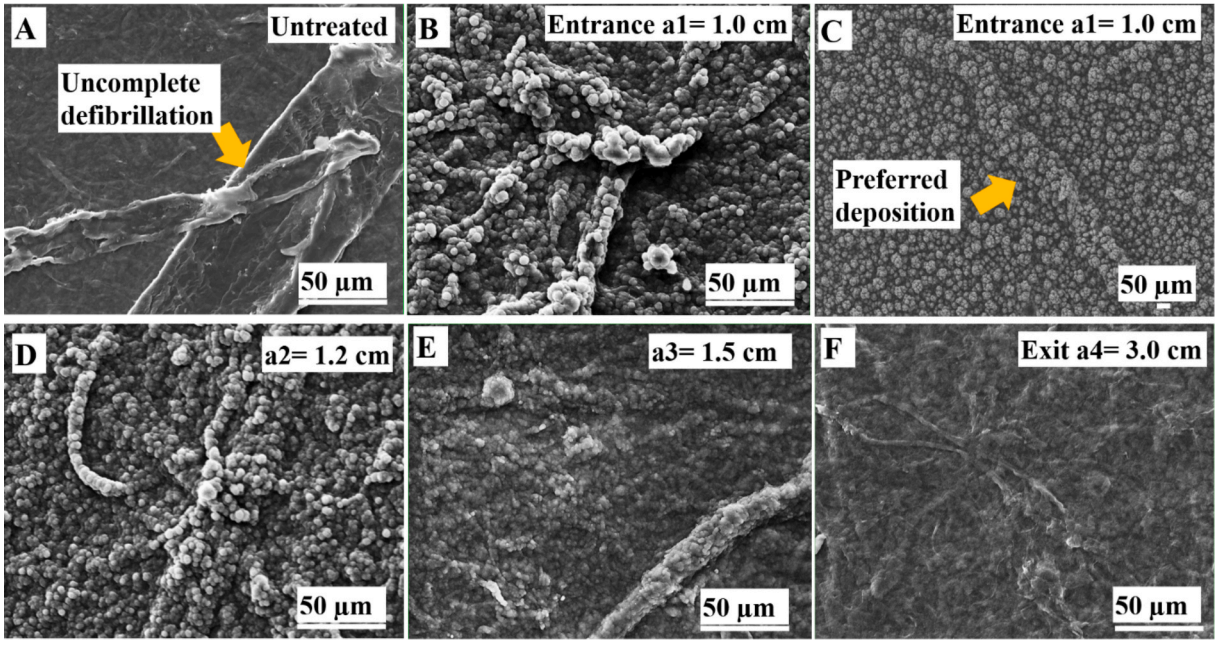


Fig. 2. Scanning Electron Microscopy (SEM) images of CNF films' surface coated by organosilicon plasma in static mode. (A) Untreated CNF films (controls), plasma-treated CNF films functionalized at (B, C) 1.0 cm, (D) 1.2 cm, (E) 1.5 cm and (F) 3.0 cm away from the entrance of the reactor. Magnification: ×1000.

roughness of 77 ± 8 nm, slightly higher than the neat films' roughness (Table 1).

The non-uniform plasma discharge can result in uneven energy distribution across the reactor [37], in turn leading to the generation of different organosilicon species as well as different concentrations of fragments throughout the reactor. For instance, Babaei et al. (2020) studied the transport phenomena of hexamethyldisiloxane (HMDSO) fragments in a glow-like discharge at atmospheric pressure (*i.e.*, helium DBD plasma reactor) on Kraft paper [29]. Their findings indicated clear variations of coating morphology along the gas flow lines which affected the water contact angle of the cellulosic substrate. These findings suggest that the observed changes in CNF films' morphology obtained in a homogeneous Townsend discharge (*i.e.,* N_2) resemble the ones previously observed with a glow-like discharge (*i.e.*, He) despite the different energies involved. Further studies are necessary to compare both physical regimes and confirm whether variations in energy distribution is a determinant factor in the observed changes in surface properties of the CNF films.

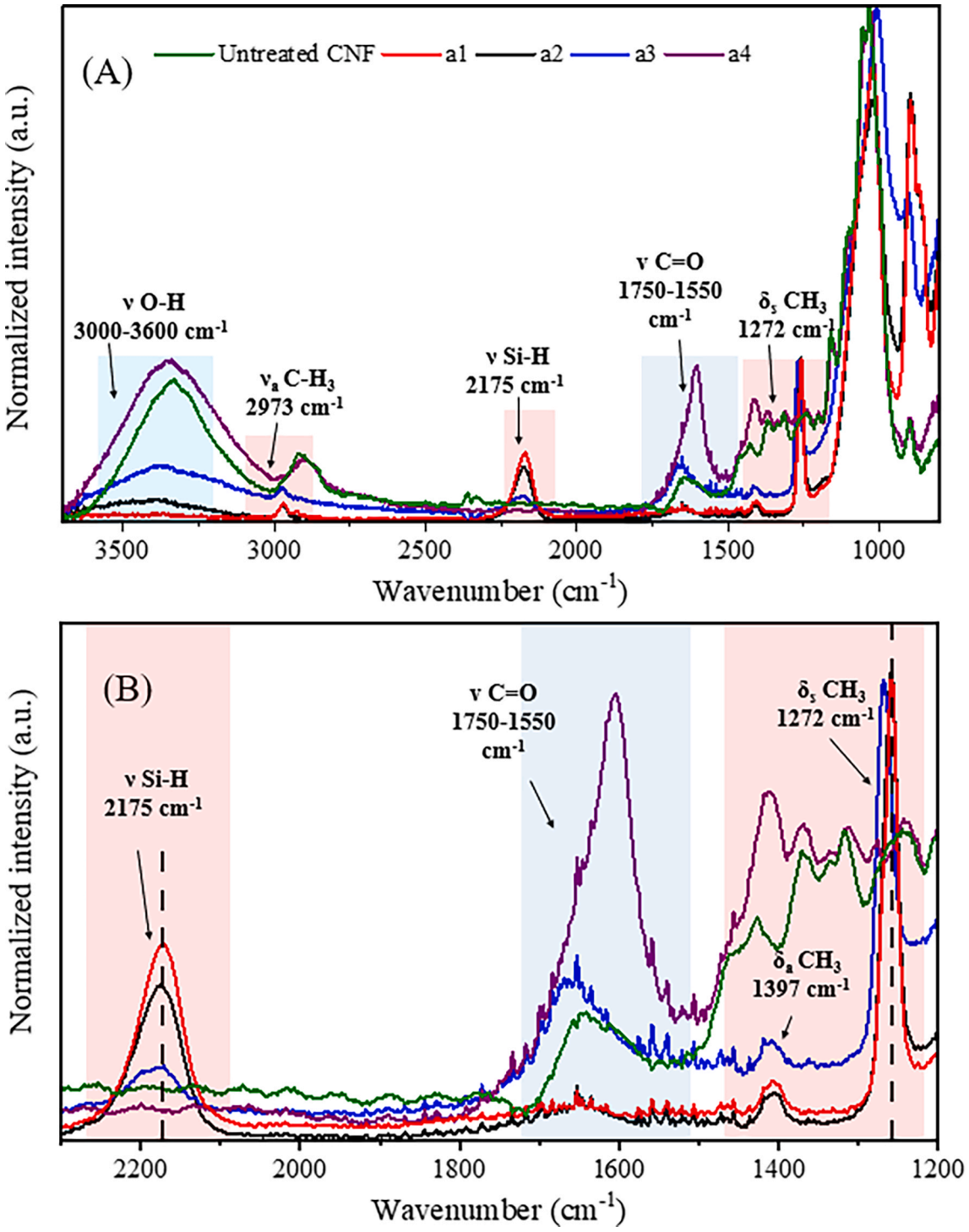


Fig. 3. Infrared (IR) spectra of untreated CNF films and films treated by organosilicon plasma in a1, a2, a3, and a4 positions (*i.e.*, 1 cm, ~1.2 cm, 1.5 cm, and 3 cm away from the entrance of the reactor) between (A) 3600-800 cm^{-1} and (B) zoomed area from 2100 to 1200 cm^{-1}.

3.2. Influence of plasma-induced fragmentation of TMCTS on CNF films' surface properties

With the observed differences in the films' surface topography across the reactor's length, variations in the films' surface chemistry and chemical functional groups distribution were investigated to identify the plasma conditions that would enable enhanced barrier performance of the films.

Fig. 3 shows the infrared spectra of CNF films before and after plasma functionalization at the different a1 to a4 positions.

Compared to the spectrum of the untreated CNF films, the spectrum of the films that were functionalized near to the entrance of the plasma reactor (*i.e.*, positions a1 and a2, Fig. 3) shows the absence of the O–H vibration signal in the 3000–3600 cm^{-1} range. Additionally, the appearance of new peaks at 2175 cm^{-1} assigned to Si-H stretching, and at 1272 cm^{-1} assigned to Si-CH_3 bending, indicates the deposition of an organosilicon layer onto the CNF films' surface [26,38] at the a1 and a2 positions. The thickness of the organosilicon layer is estimated to be higher than 0.17 μm when considering the depth of the IR analysis in this wavenumber range (3000–3600 cm^{-1}). For the films treated further away from the reactor's entrance (a3 and a4 positions), the intensity of the signal in the 3000–3600 cm^{-1} range gradually increased, suggesting the formation of silanol groups (Si-OH) or the decrease in thickness of the organosilicon coating. Silanol groups can be formed from the reaction of organosilicon fragments with water molecules from the air moisture [39,40]. The intensity of absorption of Si-H stretching observed at 2175 cm^{-1} (Fig. 3**B**) is indicative of the degree of fragmentation of TMCTS [26]. Notably, higher absorption levels at 2175 cm^{-1} were recorded near to the reactor entrance (a1 and a2 , Fig. 3**B**), suggesting the deposition of higher molecular weight fragments at the entrance of the plasma reactor than at its exit, which exhibited lower signal intensities to no detected signal (a3 and a4, Fig. 3**B**). On the other hand, the absence of the Si-H stretching signal in the IR spectrum of CNF films treated in a4 can be assigned to a higher fragmentation intensity of TMCTS, suggesting that low molecular weight fragments are evidenced at the reactor exit.

Fig. 3**B** shows two additional prominent absorbances at 1272 cm^{-1} and 1397 cm^{-1}, which are characteristic of the symmetric and asymmetric bending of the organic part (-CH_3 groups) inherent to the TCMTS chemical structure [41]. The intensity of these two signals is similar for the films treated in positions a1 and a2, suggesting that the deposition of non-fragmented TMCTS is more predominant at the entrance of the reactor than at its exit since a lower hydrocarbon content was observed in the a3 and a4 IR spectra (Fig. 3). These findings are supported by Zhang et al. (2004), who investigated the mechanism of fragmentation of organosilicon coatings in radiofrequency plasma across different discharge regions. Their studies revealed that at 5 and 10 cm downstream from the discharge region, *i.e.,* away from the reactor's entrance, no hydrocarbons were detectable in the silicon dioxide films' IR spectra. Additionally, in our study, new absorption signals were found at positions a3 and a4 (Fig. 3**B**) within the 1550–1750 cm^{-1} range, most likely arising from C=O stretching attributed to carbonyl-containing functionalities such as carboxyl (COOH), ester (COOR), or amide (CONH) groups [42]. Furthermore, this new peak might also indicate surface oxidation. The rise of polar groups such as silanol (Si-OH) and carbonyl functional groups (Fig. 3**B**) is expected to increase the films' wettability.

At this stage, we have shown that the polymerization process induced by DBD plasma changes the chemical nature of the thin coated organosilicon layer through different fragmentation. The different compositions of the gas phase are potentially linked to different distributions in energetic species between the entrance and the exit of the discharge. The explanation of the obtained results with lower TMCTS fragmentation (*i.e.*, higher hydrocarbon content) close to the reactor entrance can be related to the modification of the power discharge-to-fragment flow ratio (W/F), which varies as a function of the position in the plasma reactor. According to Petersen et al. (2011), a lower W/F ratio tends to generate organosilicon coating polydimethylsiloxane-like ($Si(CH_3)_2O$) structures, while a higher W/F ratio would reduce the organic portion of the precursor and favor the formation of silica-like (SiOx) structures [43]. Transitions to higher surface power values along the gas flow in the reactor have been observed in similar systems with DBD and nitrogen-based discharges [37]. In the present case, we hypothesized that the W/F ratio exhibits lower values at the reactor entrance, which suggests that the gradient observed in the chemical structure of the CNF films throughout the reactor can be attributed to variations in plasma discharge, and more specifically to variations in reactor's power.

3.3. Wetting behavior of plasma-treated CNF films

The introduction of new chemical functionalities on the surface of the CNF films through heterogeneous plasma deposition of organosilicon fragments with different chemistries is expected to endow the films' surface with different degrees of wettability. As highlighted in the literature, the wettability of plasma-treated surfaces depends on the degree of precursor's fragmentation [44]. Fig. 4 shows the contour plot of the water contact angle (WCA) values measured across the surface area of the plasma-treated films (from a0 to a4). The *x*-axis indicates the length of the reactor, while the *y*-axis represents the distance from the entrance to the exit of the reactor.

The WCA values varied significantly across the treated surface area: from 120° (a1 position, 1 cm away from the reactor's entrance) to ~34° at the reactor's exit (a4 position, 3 cm away from the reactor's entrance) (Fig. 4). The predominance of organic content (-CH_3) near the entrance of the reactor, as indicated by the IR spectrum (Fig. 3**B**), correlates with the hydrophobic nature of the films observed at a1. This same region also exhibited the highest roughness value ($R_a = 171 \pm 16$ nm, Table 1), which likely contributed to the increased hydrophobicity [45]. In contrast, the increase in polar groups such as silanol (Si-OH) and carbonyl functional groups at the reactor's exit correlates with the observed decrease in WCA (Fig. 4, **position a4**). The results obtained from the plasma-induced deposition of organosilicon in static mode suggest that one can potentially increase the films' overall hydrophobicity as well as their barrier properties using a dynamic deposition mode that serves to reproduce the hydrocarbon content of the areas closer to the reactor entrance, but throughout the films' surface.

3.4. Plasma-induced organosilicon deposition in dynamic mode

To enhance the homogeneity of the coating across the surface area of the films, plasma deposition of the organosilicon precursor was carried out in dynamic mode for 30 min. The parameters were adjusted to achieve the highest WCA value determined previously (*i.e.*, at the position a1, 1 cm away from the entrance of the reactor). This new approach also aims to facilitate large-scale industrial applications of DBD plasma for the functionalization of *e.g.,* roll-to-roll manufactured CNF films. Fig. 5 shows the SEM images of the surface of the films after 30 min of plasma functionalization in dynamic mode using TMCTS as the precursor and nitrogen as the carrier gas (TMCTS/N_2).

Irrespective of the treatment location (*i.e.*, a1, a3 or a4), a few macroscale fibers can still be identified on the films' surface after plasma functionalization. Compared to the films treated in static mode, the surface of the CNF films functionalized in dynamic mode were all fully covered with a uniform cauliflower-like texture, very similar to that of the films' surface treated in a1 position in static mode (*see* Fig. 2). This uniform coverage was characterized by a high WCA value of 150° (± 2°) across all three positions between the entrance and the exit of the plasma reactor (Fig. 5, **Inserts**). Compared to the untreated CNF films, which exhibited a WCA of 44° (± 3°), the organosilicon-functionalized CNF films demonstrated a superhydrophobic surface. Additionally, the WCA value obtained in dynamic mode was much higher than the WCA value obtained in static mode (150° vs. 120°), highlighting the advantages of

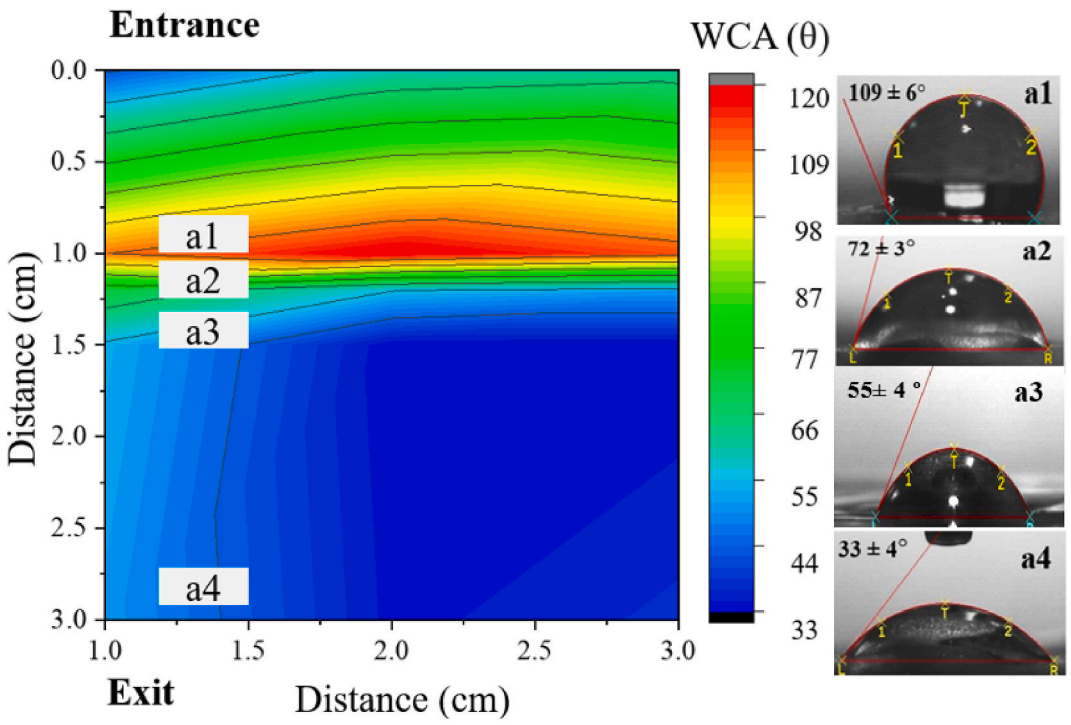


Fig. 4. Contour plot of the water contact angle values recorded across the surface area of the plasma-treated CNF films (3×3 cm^2). The top left corner of the plot (y=0) indicates the entrance of the DBD reactor and the bottom left corner (y=3 cm), the exit of the reactor . The images of the water contact angle recorded at the position a1, a2, a3 and a4 (*i.e.*, 1 cm, ~1.2 cm, 1.5 cm and 3 cm away from the entrance of the reactor) are shown on the right of the contour plot.

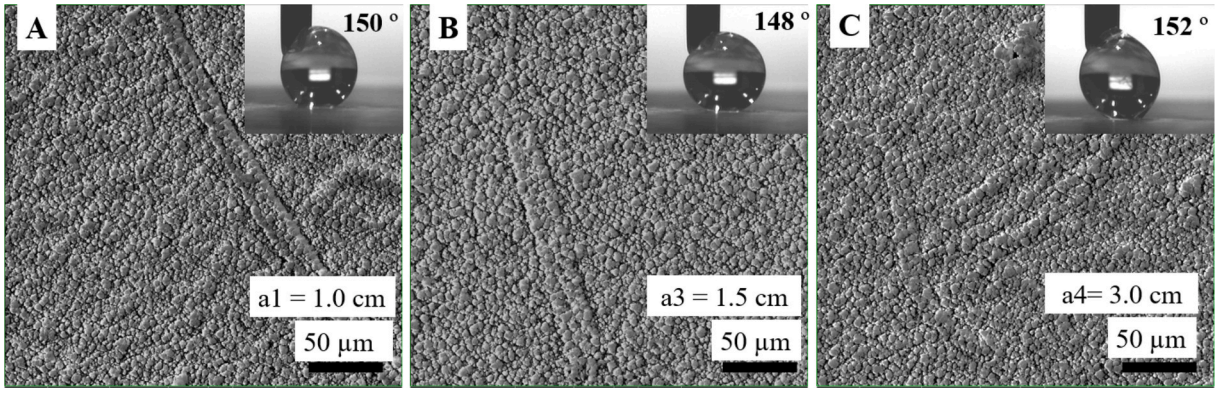


Fig. 5. SEM images of CNF films' surface after plasma functionalization in dynamic mode between the entrance and the exit of the reactor. Position (A) a1, 1 cm away from the reactor's entrance, (B) a3, 1.5 cm away, and (C) a4, 3 cm away. Magnification: ×500. Inserts show the image of a water drop on the plasma-treated films' surface at positions a1, a3, and a4 with the respective measured contact angle values.

the dynamic approach for films' functionalization. Traditional chemical approaches, such as silylation, have also enabled the creation of superhydrophobic surfaces. For instance, the reaction of TEMPO-oxidized CNFs with a polysiloxane resulted in a high WCA value of 159.6° [46]. This method, however, involved multiple reaction steps and a total reaction time of 40 min to enable uniform silylation of the films' surface. In contrast, plasma functionalization enabled uniform coverage of the films with organosilicon fragments in a single step within 30 minutes. With further optimization, this processing time could be reduced to meet the throughput demands of large-scale industrial manufacturing.

3.5. *Surface chemistry of CNF films functionalized in dynamic mode*

FTIR and XPS analyses were conducted to provide insights into the surface chemistry of functionalized CNF films and reveal the composition and chemical states of their topmost layer after plasma treatment in dynamic mode. Because the films' morphology (Fig. 5) resembled the morphology of the CNF films treated in position a1 in static mode (Fig. 2), we expected the surface chemistry of both systems to be similar.

The IR spectrum of the CNF films functionalized in dynamic mode ("Dynamic layer" label, Fig. 6**A**) shows predominant peaks that are characteristic of the organosilicon coating. Specifically, the Si-H and Si-CH_3 stretching at 2175 cm^{-1} and 1272 cm^{-1}, respectively, can be observed. These bands also appear in the IR spectrum of the films treated in static mode in position a1 (Fig. 6**A**). The absorbance of these two peaks is slightly higher after functionalization in dynamic mode than in static mode, suggesting the deposition of a thicker layer of TMCTS fragments with a higher hydrocarbon content in dynamic mode. Such a difference can be explained by the extended plasma treatment time that was used in dynamic mode, *i.e.*, 30 min (vs. 10 min in static mode).

Accordingly, the deconvoluted areas of the C1s XPS spectrum were primarily attributed to the Si-C bonds at 284.4 eV (Figs. 6**B** and **S4**). The

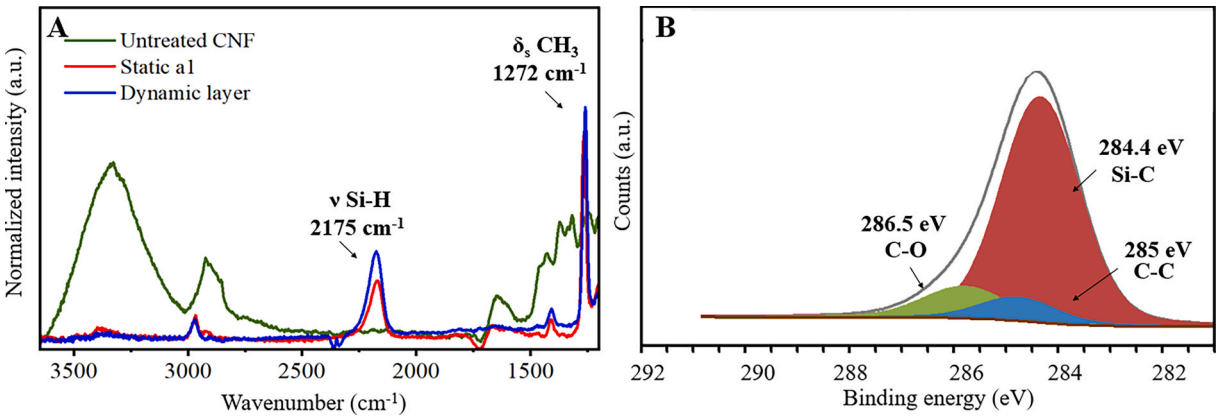


Fig. 6. (A) Infrared spectra (FTIR-ATR) of the untreated CNF films (green line, untreated CNF), the CNF films treated in static mode at the position a1 (red line, static a1), and the films treated in dynamic mode (blue line, dynamic layer). (B) Deconvoluted Carbon 1s (C1s) photoelectron spectrum (XPS) of CNF films after organosilicon deposition by plasma in dynamic mode.

signals at 286.5 eV and 285 eV can be attributed to the C-O binding energy of cellulose and the C-C binding energy of cellulose and TMCTS coating, respectively. They represent 12.1% and 7.2% of the surface chemical composition of the films, respectively (Table 2). The presence of residual C-O suggests a low probability of C-O bond formation through molecular recombination and/or partial surface coverage of the films. Given the thickness of the deposited layer (>0.17 μm), this partial coverage may be attributed to cracks caused by mechanical stresses induced during film handling. Yet, in the dynamic mode, the C–O contribution was 7.2% compared to 15.5% at a1=1 cm in static mode (**Fig. S4**), indicating better organosilicon coverage in the dynamic mode. The presence of molecular defects in the organosilicon coating did not impact the development of a superhydrophobic film surface.

The analysis of the CNF films' surface morphology and chemistry after plasma treatment in dynamic mode confirms that the target conditions and performance identified in static mode (here, a high water contact angle value) can be reproduced in dynamic conditions with an extended plasma treatment time. Compared to the functionalization in static mode, the dynamic approach enabled a uniform deposition of organosilicon fragments on the surface of the CNF films, which in turn resulted in a uniform hydrophobization of the films' surface. Such a result is expected to contribute to the moisture and water barrier performance of the CNF films; yet the small coating cavities that were observed by XPS may be large enough to facilitate moisture diffusion through the films and in turn, weaken the films' barrier performance.

Hence, we adopted a multi-layer approach to counteract the presence of molecular defects (or holes) in the single plasma-coated layer of organosilicon fragments. The stacking of multi-layers of different barrier performance is commonly exploited for the manufacturing of high-barrier flexible packaging substrates [47]. Thus, this strategy has been implemented to further develop the barrier performance of the CNF films.

3.6. *Tuning the wettability of CNF films by multilayer plasma treatment*

The wettability and surface chemistry of the CNF films treated in dynamic mode using TMCTS/N_2 plasma were tuned by altering the composition of the plasma gas from TMCTS/N_2 to N_2O/N_2 to modify the nature of the functional groups of the coated organosilicon fragments. The use of nitrous oxide gas (N_2O) aims to oxidize the TMCTS-derived plasma layer and remove the hydrophobic functional groups (such as -CH_3) from the coated fragments [22]. This second plasma treatment (or second layer) was applied on the surface of the already-plasma-treated CNF films (with TMCTS/N_2) to create a more inorganic denser layer and increase the film's wettability. A third layer was then conducted using TMCTS/N_2 plasma (composition similar to that of the first layer) on the films' surface, right after the N_2O/N_2 plasma treatment to create an upper hydrophobic layer and enclose the hydrophilic second layer between two chemically different surfaces. The second plasma treatment (*i.e.,* N_2O/N_2) also served as an intermediate layer, aiming to improve the uniformity and coverage of the organosilicon coating onto the CNF film's surface. We hypothesized that such a thin, hydrophilic layer could help prevent potential delamination of successive organosilicon layers, which was observed and reported by Profili et al. (2020). Fig. 7 shows the SEM images of the CNF films' surface after 1 min of plasma functionalization in dynamic mode using the oxidizing gas mixture N_2O/N_2 (*i.e.*, second layer, Fig. 8A) and after organosilicon coating using TMCTS/N_2 for 30 min (*i.e.*, third layer, Fig. 7**B**).

The SEM image of the second plasma layer does not highlight a strong modification of the coating morphology (Fig. 7**A**), suggesting that the oxidation process did not affect the morphology of the previous plasma layer (first layer). However, this second plasma treatment did induce apparent structural defects on the films' surface that can be related to physical etching (Fig. 7**A, Insert**). This phenomenon can be attributed to changes in the physical regime of the discharge [48], potentially leading to instabilities such as those observed in a filamentary regime. Filamentary discharges are usually observed at atmospheric discharges, specifically when an oxidizing gas and/or a complex surface are used in a DBD configuration at atmospheric pressure. They have been also observed in previous studies [21]. As discussed in the literature, the presence of defects, such as variations in the films' topography or a wavy surface, can alter the physical regime [49]. Interestingly, the deposition of the third plasma layer using TMCTS/N_2 enabled the coverage of the defects induced by the N_2O/N_2 plasma treatment (Fig. 6**B**). Similar to the first layer (see Fig. 5), the third one exhibited a cauliflower-like structure (Fig. 6**B**), which can also be attributed to the uniform deposition of organosilicon fragments onto the films' surface.

The surface roughness of the three layers was also measured by AFM (Table 3). The first and third layers obtained by TMCTS/N_2 plasma presented higher average roughness values than the second layer obtained by N_2O/N_2 plasma.

The surface chemistry of the CNF films after deposition of the second and the third layer by plasma also reveals clear distinctions. Compared to the IR spectrum of the first layer (Fig. 8**A**), the spectra of the second

Table 2
XPS percentage atomic concentrations of C1s components on the CNF films' surface after organosilicon deposition by plasma in dynamic mode.

Element	Si-C	C-C	C-O
Binding energy (eV)	284.4	285	286.5
Atomic concentration (%)	80.7	12.1	7.2

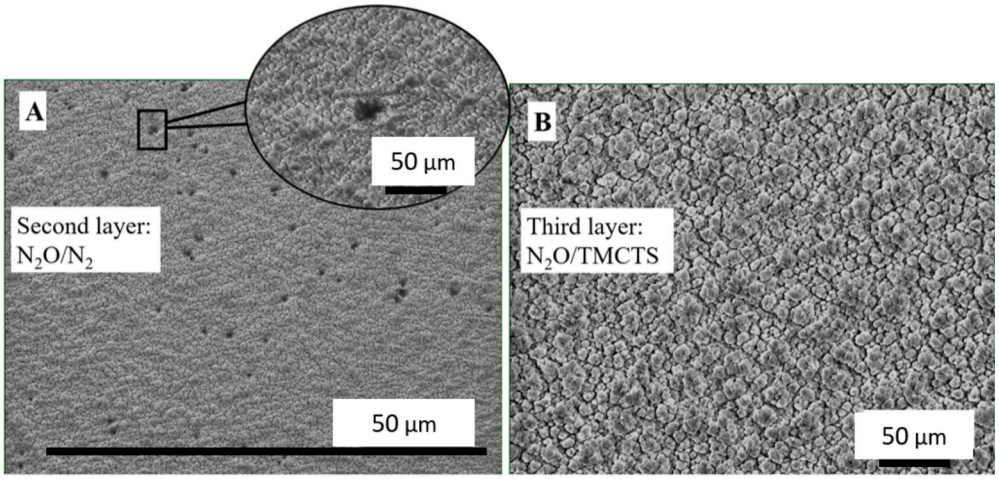


Fig. 7. SEM images of the organosilicon-coated CNF films' surface after (A) a second plasma layer using N_2O/N_2 and (B) a third plasma layer using TMCTS/N_2. Magnifications: (A) ×200 and ×500 (insert), (B) ×1000.

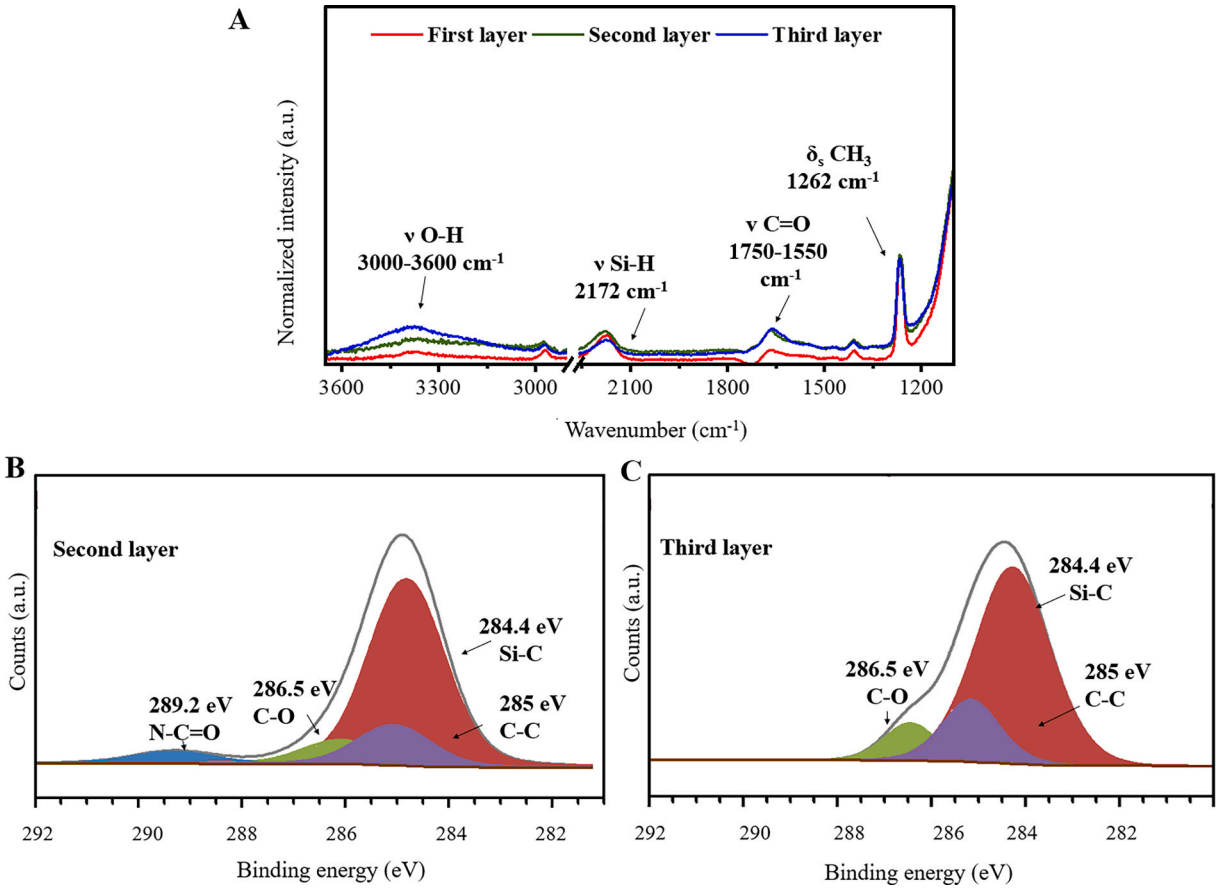


Fig. 8. Surface chemistry analysis of the CNF films after successive plasma deposition of a first layer with TMCTS/N_2, a second layer with N_2O/N_2, and a third layer with TMCTS/N_2. Infrared spectra from 3600 to 800 cm^{-1} (A) and X-ray photoelectron (XPS) carbon C1s spectra of the (B) second layer and the (C) third layer.

Table 3
Average roughness (R_a) values of the first TMCTS/N_2, the second N_2O/N_2 and the third TMCTS/N_2 plasma layers. The average roughness values were measured by Atomic Force Microscopy.

	First layer	Second layer	Third layer
Plasma treatment	TMCTS/N_2	N_2O/N_2	TMCTS/N_2
R_a (nm)	225± 9	190 ± 52	245 ± 4

and third layers show an increase of the O-H stretching signal intensity (3600–3000 cm^{-1}). In the case of the second layer, OH groups may have been created via oxidation of the film's surface (in the micrometer order) by N_2O/N_2 plasma. When analyzing the third layer, the presence of a higher number of O-H groups (than in the first layer) may be explained by a modification of the discharge and a slightly different regime, since the discharge was formed onto another surface type. In the 1550–1750 cm^{-1} range, a weaker absorption signal can also be observed on both IR spectra, which most likely indicates the presence of carbonyl-

containing functionalities [42]. The characteristic peaks of the organosilicon coating at 2172 cm^{-1} and 1262 cm^{-1} assigned to the Si-H stretching and Si-CH_3 bending, respectively, are also present on the spectrum of the three plasma layers [26,38]. Despite the presence of the second plasma layer conducted with a mixture of oxidizing gases, the signals of the organosilicon fragments were still detectable by IR and exhibited similar intensities. This suggests that (i) the second plasma treatment did not affect the fragmentation of the organosilicon precursor within the depth of analysis (0.17 μm) and (ii) the oxidation process primarily affected the topmost nanometer layer of the films [50]. In fact, the introduction of an oxidizing gas was expected to increase the oxidation intensities of the films, which was confirmed by XPS analysis, showing a 70% increase in the O/C ratio of the CNF films treated with N_2O/N_2 (Table 4 and **Fig. S5**).

The XPS analysis confirmed the formation of a slight amount of amide (N-C=O) functional groups (signal at 289.2 eV, Fig. 8**B**) on the surface of the CNF films treated by NO_2/N_2 plasma. These functional groups were likely introduced by the nitrogen from the oxidizing gas during the NO_2/N_2 treatment, as observed in previous studies [40,42]. Similarly to the observed changes in OH band intensity, the IR intensity of these bands remained high considering that no precursor was introduced at this oxidation stage, indicating that the fixation of oxygen and nitrogen moieties on the surface remained significant. At this stage, the high surface area of the coating combined to the potential penetration of reactive species in the first layer could explain the observed high concentrations in functional groups. Further chemical depth profiling is necessary to clarify the penetration depth of the plasma under such treatment conditions and the chemical processes involved over time.

The presence of carbonyl-containing and hydroxyl chemical functionalities in the third layer may be explained by the release of oxygen-based molecules trapped in the porous substrate during plasma treatment, as evidenced in the work by Levasseur et al. (2013) [51].

The wettability of the CNF films after the second and third plasma treatment is shown Fig. 9 and evidences the clear difference in surface chemistry between the second and third plasma treatment.

The untreated CNF films had an initial WCA of 47 ± 3° that gradually decreased overtime until reaching 40 ± 1° after 10 seconds (Fig. 9, *left column*). CNF films have a known hydrophilic nature due to the presence of hydroxyl groups that allow them to interact with considerable amounts of water [52].

The treatment of the films with N_2O/N_2 plasma decreased the WCA even further to an initial value of 11 ± 2° (Fig. 9, *middle column*). Within 3 seconds, the complete spreading and disappearance of the water droplet on the film's surface was observed. At this stage, the water seemed to have spread over the inorganic layer of the film and could then act as a dense barrier. However, considering the thickness of the CNF film, it remains difficult to conclude whether the water drop completely absorbed into the layer or spread. Additional analysis will be conducted to confirm the penetration depth of the water drop inside this thin plasma layer. All in all, this second oxidizing plasma treatment effectively removed the hydrophobic functional groups of the previously deposited organosilicon coating, creating a hydrophilic layer on the top of the first hydrophobic layer, as previously shown by XPS (Table 3). These observations align with the literature, where treatments involving oxidizing gases such as N_2O have been shown to modify the surface chemistry of TMCTS, enhancing the hydrophilicity of the deposited fragments (WCA<5°) through the introduction of polar functional groups [53].

Table 4
XPS percentage atomic concentrations of Carbon and Oxygen, and O/C ratio for CNF films plasma-treated with 1, 2 or 3 layers of different compositions.

	Plasma treatment	Carbon (C)%	Oxygen (O)%	O/C
First layer	TMCTS/N_2	26.56 ± 0.54	28.88 ± 0.30	1.0 ± 0
Second layer	N_2O/N_2	20.58 ± 0.81	34.74 ± 0.39	1.7 ± 0.10
Third layer	TMCTS/N_2	22.89 ± 0.82	30.42 ± 1.21	1.3 ± 0.10

The enhanced hydrophilicity of the CNF films' surface did not alter the effectiveness of the third plasma treatment using TMCTS/N_2. Indeed, the TMCTS/N_2 plasma treatment reestablished the superhydrophobicity of the CNF films' surface, showing a stable WCA value of 150° (Fig. 9, *right column*).

Because the presence of a superhydrophobic surface does not necessarily endow the bulk material with high barrier performance against water and moisture [54], we tested the water vapor transmission rate (WVTR) and water absorption capacity (WA) of the CNF films, after a first, a second, and a third plasma treatment. The results are shown in Fig. 10.

Untreated CNF films are expected to exhibit a high WVTR and a high WA due to the hydrophilic nature of cellulose and its swelling ability [4]. In this study, the untreated CNF films had a WVTR of 168 g/m^2.day and a WA capacity of 120%. After the deposition of the first plasma layer (TMCTS/N_2), the water absorption of the films drastically decreased from 120% to 0.3% because of the presence of the superhydrophobic organosilicon coating (Fig. 10). In contrast, the WVTR was reduced by 15% only. The small molecular defects that were observed by XPS may have facilitated the moisture diffusion through the films, explaining the marginal improvement of the films' moisture barrier [28].

The second plasma treatment resulted in a drastic increase of the WVTR that exceeded the performance of the untreated CNF films. Such an increase in moisture transfer may be due to the combined effect of the increased hydrophilicity of the CNF films' surface and the introduction of structural defects in the films, that were observed Fig. 7. Despite the increase in WVTR, the WA of the films remained low, demonstrating the effectiveness of the hydrophobic layers in mitigating water absorption into the cellulose substrate.

The treatment of the films with a third plasma layer (TMCTS/N_2) resulted in a WVTR of 180 g/m^2.day, which remained higher than the WVTR of the untreated CNF films. While the SEM observations suggested complete coverage of the defects induced by the second plasma treatment, nano- and sub-nano-sized pores may still be present in the films after plasma deposition of molecular fragments. The increased roughness of the third plasma layer may also have contributed to the higher WVTR values. As previously shown by Van Assche et al. (2014) [55], a high surface roughness combined with the presence of defects can lead to more accessible pathways for moisture permeation. In contrast, the WA of the CNF films treated with three layers remained low, at 1%, demonstrating the effective water-resistant nature of the coating and its ability to mitigate hydrophilicity and the action of capillary forces on water absorption.

Further studies will thus be necessary to optimize the plasma-induced multi-layer deposition of organosilicon fragments onto the CNF films to develop higher moisture barrier performance, since acting on the surface chemistry alone is not sufficient to meet moisture barrier requirements of food packaging. However, the ability of plasma to increase water vapor transmission rate of CNF films without impacting their barrier against liquid, thus the ability of plasma to create breathable bio-sourced substrates can be of further interest in other applications such as wound dressing.

4. Conclusion

This study investigates the potential of dielectric barrier discharge (DBD) plasma in tuning and tailoring the surface properties of CNF films for enhanced water and moisture barrier performance. Plasma deposition of an organosilicon precursor in static mode proved to be an effective approach to understand the preferential deposition and fragmentation process of the precursor onto the CNF films' surface between the entrance and the exit of the plasma reactor. Specifically, the static-mode approach served as a foundation to a more controlled plasma

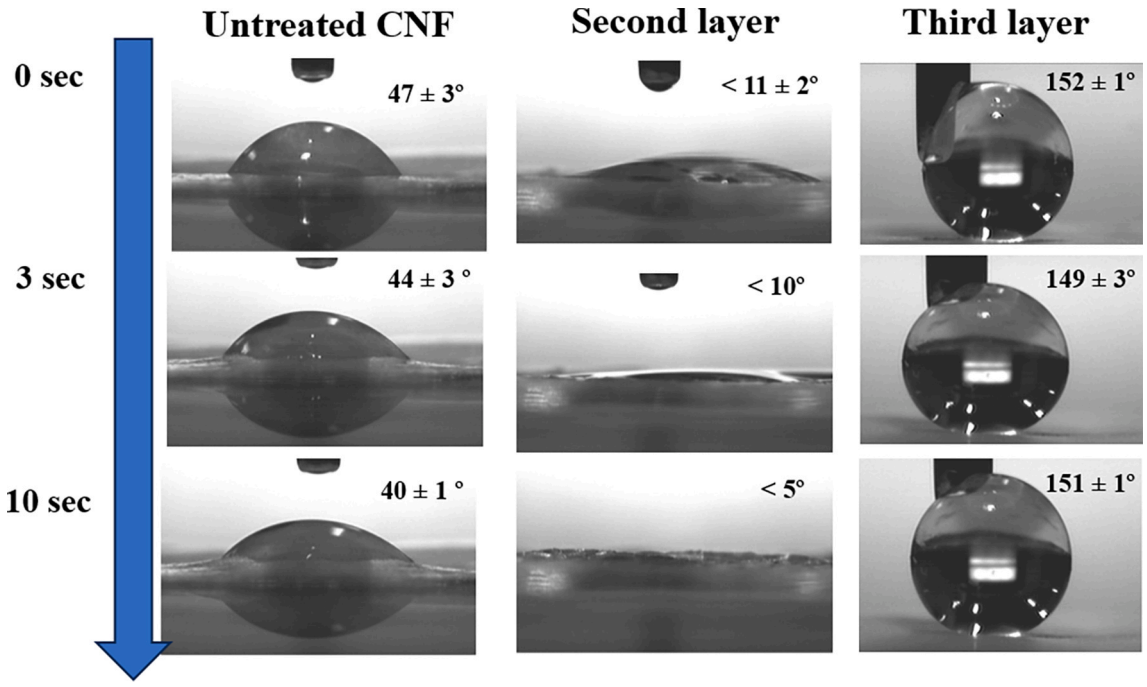


Fig. 9. Dynamic water contact angle analysis of the CNF films after 0, 3 and 10 seconds of contact with a water droplet. Untreated CNF films' surface *(Left column)*. CNF films' surface after two consecutive plasma treatments (TMCTS/N_2 and N_2O/N_2) *(middle column)*, and after three consecutive plasma treatments (TMCTS/N_2, N_2O/N_2 and TMCTS/N_2) *(right column)*.

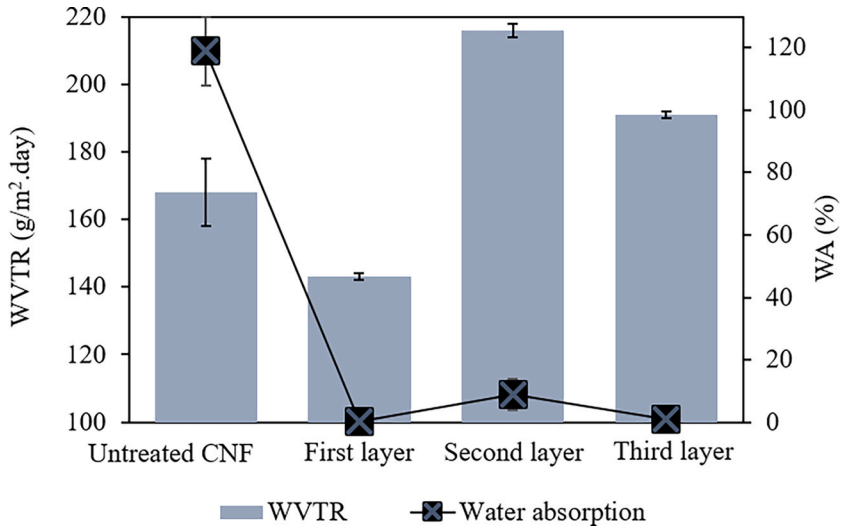


Fig. 10. Water vapor transmission rate (WVTR in g/m^2day) and water absorption capacity (WA in%) of the untreated CNF films, and films functionalized by one (TMCTS/N_2), two (layer 1 + N_2O/N_2) and three (layer 2 + TMCTS/N_2) plasma treatments.

treatment, conducted in dynamic mode for scalability, that resulted in the formation of a uniform, porous superhydrophobic coating onto the entire CNF films' surface.

Through the dynamic mode approach, a multi-layer strategy was then implemented for the finetuning of the barrier properties of the CNF films. Successive hydrophobic and hydrophilic layers were applied onto the CNF films by varying the nature of the plasma gas. The introduction of an oxidizing gas increased considerably the wettability and hydrophilicity of the previously hydrophobized CNF films by plasma-induced organosilicon coating but generated structural defects that were responsible for the marginal decrease in moisture transfer. However, this second hydrophilic layer enabled the deposition of another plasma-induced organosilicon coating that exhibited a superhydrophobic surface and a very low (<1%) water absorption capability.

The ability to tailor surfaces through plasma treatment holds promise for enhancing the water barrier of CNF films through molecular coating of organosilicon fragments and for improving the films' wettability to serve *e.g.,* as an intermediate tie layer in a multilayer packaging system. Further research is still needed to understand how the creation of structural defects by plasma can be minimized to ultimately mitigate the effect of plasma on the moisture barrier performance of CNF films and the impact on mechanical performance and degradability of the films. This work has also highlighted the crucial role of the second, intermediate plasma layer in tuning the barrier performance of the films. A simple oxidation process resulted in the creation of a highly breathable CNF films yet barrier to liquids, through changes in the films' surface

structure and penetration of reactive species in the coated films. Further research is thus needed to decipher the underlying chemical mechanisms of the oxidation process, determine the penetration depth of plasma in the coated films and understand the role of reactive species on the diffusion of gas molecules.

5. Funding sources

This research was supported in part by the College of Natural Resources at NC State University, the 2022-23 NC State Internationalization Seed Grant, the Mitacs Globalink Research Award (#IT35099) and the intramural research program of the U.S. Department of Agriculture, National Institute of Food and Agriculture, McIntire-Stennis, grant number NI22MSCFRXXXG054. These findings and conclusions have not been formally disseminated by the U. S. Department of Agriculture and Should not be construed to represent any agency determination or policy.

CRediT authorship contribution statement

Mirela A. Artner: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Lucia Švandová:** Visualization, Software, Methodology, Data curation. **Nathalie Lavoine:** Formal analysis, Methodology, Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Jacopo Profili:** Writing – review & editing, Supervision, Resources, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mirela Artner reports financial support was provided by Mitacs Inc. Nathalie Lavoine reports financial support was provided by National Institute of Food and Agriculture. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Dr. Alex Destrieux and Dr. Williams Caceres Ferreira for their explanations and guidance on the dielectric barrier discharge plasma device, and Mrs. Andrée-Anne Guay-Bégin for conducting the XPS analyses.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsusc.2026.168134.

Data availability

Data will be made available on request.

References

[1] U.S. Plastic Pact: 2021 Annual report, U.S. Plastics Pact: 2021 Annual report, 2021. https://usplasticspact.org/2021-report/ (accessed November 16, 2023).
[2] S.L. Fávaro, A.R. Freitas, T.A. Ganzerli, A.G.B. Pereira, A.L. Cardozo, O. Baron, E. C. Muniz, E.M. Girotto, E. Radovanovic, PET and aluminum recycling from multilayer food packaging using supercritical ethanol, J. Supercrit. Fluids 75 (2013) 138–143, https://doi.org/10.1016/j.supflu.2012.12.015.
[3] A.S. Bauer, M. Tacker, I. Uysal-Unalan, R.M.S. Cruz, T. Varzakas, V. Krauter, Recyclability and redesign challenges in multilayer flexible food packaging—a review, Foods 10 (2021), https://doi.org/10.3390/foods10112702.
[4] E. Pasquier, B.D. Mattos, H. Koivula, A. Khakalo, M.N. Belgacem, O.J. Rojas, J. Bras, Multilayers of renewable nanostructured materials with high oxygen and water vapor barriers for food packaging, ACS Appl. Mater. Interfaces 14 (2022) 30236–30245, https://doi.org/10.1021/acsami.2c07579.
[5] Q. Meng, T.J. Wang, Mechanics of strong and tough cellulose nanopaper, Appl. Mech. Rev. 71 (2019) 1–30, https://doi.org/10.1115/1.4044018.
[6] Q. Tarrés, H. Oliver-Ortega, P.J. Ferreira, M. Àngels Pèlach, P. Mutjé, M. Delgado-Aguilar, Towards a new generation of functional fiber-based packaging: cellulose nanofibers for improved barrier, mechanical and surface properties, Cellul. 25 (2018) 683–695, https://doi.org/10.1007/s10570-017-1572-7.
[7] G. Fotie, S. Limbo, L. Piergiovanni, Manufacturing of food packaging based on nanocellulose: current advances and challenges, Nanomaterials 10 (2020) 1–26, https://doi.org/10.3390/nano10091726.
[8] M.D. Steven, J.H. Hotchkiss, Comparison of flat film to total package water vapour transmission rates for several commercial food wraps, Packag. Technol. Sci. 15 (2002) 17–27, https://doi.org/10.1002/pts.562.
[9] D. Cheng, P. Wei, L. Zhang, J. Cai, Surface-initiated atom transfer radical polymerization grafting from nanoporous cellulose gels to create hydrophobic nanocomposites, RSC Adv. 8 (2018) 27045–27053, https://doi.org/10.1039/c8ra04163f.
[10] H. Sehaqui, T. Zimmermann, P. Tingaut, Hydrophobic cellulose nanopaper through a mild esterification procedure, Cellul. 21 (2014) 367–382, https://doi.org/10.1007/s10570-013-0110-5.
[11] W. Li, S. Wang, W. Wang, C. Qin, M. Wu, Facile preparation of reactive hydrophobic cellulose nanofibril film for reducing water vapor permeability (WVP) in packaging applications, Cellul. 26 (2019) 3271–3284, https://doi.org/10.1007/s10570-019-02270-x.
[12] M. Andresen, L.S. Johansson, B.S. Tanem, P. Stenius, Properties and characterization of hydrophobized microfibrillated cellulose, Cellul. 13 (2006) 665–677, https://doi.org/10.1007/s10570-006-9072-1.
[13] G. Rodionova, M. Lenes, Ø. Eriksen, Ø. Gregersen, Surface chemical modification of microfibrillated cellulose: improvement of barrier properties for packaging applications, Cellul. 18 (2011) 127–134, https://doi.org/10.1007/s10570-010-9474-y.
[14] M.A. Poothanari, Y. Leterrier, Fluorine free surface modification of microfibrillated cellulose-clay composite films: effect of hydrophobicity on gas barrier performance, Surfaces 7 (2024) 283–295, https://doi.org/10.3390/surfaces7020019.
[15] W. Wang, F. Gu, Z. Deng, Y. Zhu, J. Zhu, T. Guo, J. Song, H. Xiao, Multilayer surface construction for enhancing barrier properties of cellulose-based packaging, Carbohydr. Polym. 255 (2021), https://doi.org/10.1016/j.carbpol.2020.117431.
[16] M. Jonoobi, J. Harun, A.P. Mathew, M.Z.B. Hussein, K. Oksman, Preparation of cellulose nanofibers with hydrophobic surface characteristics, Cellul. 17 (2010) 299–307, https://doi.org/10.1007/s10570-009-9387-9.
[17] M. Henriksson, L.A. Berglund, P. Isaksson, T. Lindström, T. Nishino, Cellulose nanopaper structures of high toughness, Biomacromolecules 9 (2008) 1579–1585, https://doi.org/10.1021/bm800038n.
[18] B.P. Frank, D.P. Durkin, E.R. Caudill, L. Zhu, D.H. White, M.L. Curry, J.A. Pedersen, D.H. Fairbrother, Impact of silanization on the structure, dispersion properties, and biodegradability of nanocellulose as a nanocomposite filler, ACS Appl. Nano Mater. 1 (2018) 7025–7038, https://doi.org/10.1021/acsanm.8b01819.
[19] M. Fazeli, J.P. Florez, R.A. Simão, Improvement in adhesion of cellulose fibers to the thermoplastic starch matrix by plasma treatment modification, Compos. B Eng. 163 (2019) 207–216, https://doi.org/10.1016/j.compositesb.2018.11.048.
[20] S. Kakkar, B. Harjani, N. Ledwani, L. Ledwani, Nanotechnology for Energy and Environmental Engineering, 2020. http://link.springer.com/10.1007/978-3-030-33774-2_2%0Ahttp://link.springer.com/10.1007/978-3-030-33774-2.
[21] J. Profili, S. Asadollahi, P. Vinchon, A. Dorris, S. Beck, A. Sarkassian, L. Stafford, Recent progress on organosilicon coatings deposited on bleached unrefined Kraft paper by non-thermal plasma process at atmospheric pressure, Prog. Org. Coat. 147 (2020), https://doi.org/10.1016/j.porgcoat.2020.105865.
[22] L. Maechler, C. Sarra-Bournet, P. Chevallier, N. Gherardi, G. Laroche, Anti-fog layer deposition onto polymer materials: a multi-step approach, Plasma Chem. Plasma Process. 31 (2011) 175–187, https://doi.org/10.1007/s11090-010-9261-4.
[23] A. Tóth, L. Černáková, M. Černák, K. Kunovská, Surface analysis of groundwood paper treated by diffuse coplanar surface barrier discharge (DCSBD) type atmospheric plasma in air and in nitrogen, Holzforschung 61 (2007) 528–531, https://doi.org/10.1515/HF.2007.080.
[24] L.C. Vander Wielen, M. Östenson, P. Gatenholm, A.J. Ragauskas, Surface modification of cellulosic fibers using dielectric-barrier discharge, Carbohydr. Polym. 65 (2006) 179–184, https://doi.org/10.1016/j.carbpol.2005.12.040.
[25] M. Theelen, D. Habets, L. Staemmler, H. Winands, P. Bolt, Localised plasma deposition of organosilicon layers on polymer substrates, Surf. Coat. Technol. 211 (2012) 9–13, https://doi.org/10.1016/j.surfcoat.2011.10.022.
[26] J. Zhang, D.S. Wavhal, E.R. Fisher, Mechanisms of SiO2 film deposition from tetramethylcyclotetrasiloxane, dimethyldimethoxysilane, and trimethylsilane plasmas, J. Vac. Sci. Technol. A 22 (2004) 201–213, https://doi.org/10.1116/1.1635392.
[27] J. Profili, S. Babaei, M. Al Rashidi, A. Dorris, S. Asadollahi, A. Sarkissian, L. Stafford, Plasma-deposited organosilicon hydrophobic coatings on cellulosic materials for wet packaging applications, Coatings 13 (2023), https://doi.org/10.3390/coatings13050924.
[28] S. Babaei, J. Profili, M. Al Rashidi, A. Dorris, S. Beck, S. Asadollahi, A. Sarkissian, L. Stafford, Permeation properties of a plasma-processed organosilicon-carboxymethylcellulose bilayer on fibrillated cellulosic films for sustainable packaging applications, Cellul. (2023), https://doi.org/10.1007/s10570-023-05367-6.

[29] S. Babaei, J. Profili, S. Asadollahi, A. Sarkassian, A. Dorris, S. Beck, L. Stafford, Analysis of transport phenomena during plasma deposition of hydrophobic coatings on porous cellulosic substrates in plane-to-plane dielectric barrier discharges at atmospheric pressure, Plasma Processes Polym. 17 (2020), https://doi.org/10.1002/ppap.202000091.
[30] J. Vartiainen, T. Malm, Surface hydrophobization of CNF films by roll-to-roll HMDSO plasma deposition, J. Coat. Technol. Res. 13 (2016) 1145–1149, https://doi.org/10.1007/s11998-016-9833-1.
[31] A. Khakalo, T. Mäkelä, L.S. Johansson, H. Orelma, T. Tammelin, High-throughput tailoring of nanocellulose films: from complex bio-based materials to defined multifunctional architectures, ACS Appl. Bio Mater. 3 (2020) 7428–7438, https://doi.org/10.1021/acsabm.0c00576.
[32] S.H. Osong, S. Norgren, P. Engstrand, Processing of wood-based microfibrillated cellulose and nanofibrillated cellulose, and applications relating to papermaking: a review, Cellul. 23 (2016) 93–123, https://doi.org/10.1007/s10570-015-0798-5.
[33] T 248 sp-00, T 248 sp-00 Laboratory beating of pulp (PFI mill method), 2000.
[34] ASTM E96, Standard Test Methods for Water Vapor Transmission of Materials, 2000. www.astm.org.
[35] TAPPI 441 om-09, T441- Water absorption, 2013.
[36] H.J. Kim, S. Roy, J.W. Rhim, Effects of various types of cellulose nanofibers on the physical properties of the CNF-based films, J. Environ. Chem. Eng. 9 (2021) 106043, https://doi.org/10.1016/j.jece.2021.106043.
[37] C. Tyl, X. Lin, M.C. Bouzidi, S. Dap, H. Caquineau, P. Ségur, N. Gherardi, N. Naudé, Investigation of memory effect in atmospheric pressure dielectric barrier discharge in nitrogen with small oxygen or nitric oxide addition, J. Phys. D Appl. Phys. 51 (2018), https://doi.org/10.1088/1361-6463/aad472.
[38] E. Todd Ryan, S.M. Gates, S.A. Cohen, Y. Ostrovski, E. Adams, K. Virwani, A. Grill, Effect of low-frequency radio frequency on plasma-enhanced chemical vapor deposited ultra low-κ dielectric films for very large-scale integrated interconnects, J Appl Phys 115 (2014), https://doi.org/10.1063/1.4870453.
[39] P.P. Joshi, R. Pulikollu, S.R. Higgins, X. Hu, S.M. Mukhopadhyay, Investigation of growth, coverage and effectiveness of plasma assisted nano-films of fluorocarbon, Appl. Surf. Sci. 252 (2006) 5676–5686, https://doi.org/10.1016/j.apsusc.2005.07.061.
[40] I. Rodríguez Durán, A. Durocher-Jean, J. Profili, L. Stafford, G. Laroche, Atmospheric pressure Townsend discharges as a promising tool for the one-step deposition of antifogging coatings from N2O/TMCTS mixtures, Plasma Processes Polym. 17 (2020), https://doi.org/10.1002/ppap.201900186.
[41] D. Barreca, A. Gasparotto, C. Maccato, C. Maragno, E. Tondello, G. Rossetto, Low-temperature PECVD of transparent SiOxCyH z thin films, Chem. Vap. Depos. 13 (2007) 205–210, https://doi.org/10.1002/cvde.200606518.
[42] R. Reuter, N. Gherardi, J. Benedikt, Effect of N 2 dielectric barrier discharge treatment on the composition of very thin SiO 2-like films deposited from hexamethyldisiloxane at atmospheric pressure, Appl. Phys. Lett. 101 (2012), https://doi.org/10.1063/1.4764938.
[43] J. Petersen, R. Bechara, J. Bardon, T. Fouquet, F. Ziarelli, L. Daheron, V. Ball, V. Toniazzo, M. Michel, A. Dinia, D. Ruch, Atmospheric plasma deposition process: a versatile tool for the design of tunable siloxanes-based plasma polymer films, Plasma Processes Polym. 8 (2011) 895–903, https://doi.org/10.1002/ppap.201100022.
[44] S. Asadollahi, J. Profili, M. Farzaneh, L. Stafford, Development of organosilicon-based superhydrophobic coatings through atmospheric pressure plasma polymerization of HMDSO in nitrogen plasma, Materials 12 (2019), https://doi.org/10.3390/ma12020219.
[45] D.M. Spori, T. Drobek, S. Zürcher, M. Ochsner, C. Sprecher, A. Mühlebach, N. D. Spencer, Beyond the lotus effect: roughness influences on wetting over a wide surface-energy range, Langmuir 24 (2008) 5411–5417, https://doi.org/10.1021/la800215r.
[46] S. Chen, Y. Song, F. Xu, Highly transparent and hazy cellulose nanopaper simultaneously with a self-cleaning superhydrophobic surface, ACS Sustain. Chem. Eng. 6 (2018) 5173–5181, https://doi.org/10.1021/acssuschemeng.7b04814.
[47] J. Fahlteich, S. Mogck, T. Wanski, N. Schiller, S. Amerg-Shwab, U. Weber, O. Miesbauer, E. Kücükpinar-Niarchos, K. Noller, C. Boeffel, The role of defects in single- and multi-layer barriers for flexible electronics, in: Society of Vacuum Coaters 57th (2014) Annual Technical Conference Proceedings, Society of Vacuum Coaters, 2015: pp. 403-413. Doi: 10.14332/svc14.proc.1881.
[48] D. Merche, N. Vandencasteele, F. Reniers, Atmospheric plasmas for thin film deposition: a critical review, Thin Solid Films 520 (2012) 4219–4236, https://doi.org/10.1016/j.tsf.2012.01.026.
[49] P. Rajasekaran, P. Mertmann, N. Bibinov, D. Wandke, W. Viöl, P. Awakowicz, Filamentary and homogeneous modes of dielectric barrier discharge (DBD) in air: Investigation through plasma characterization and simulation of surface irradiation, Plasma Processes Polym. 7 (2010) 665–675, https://doi.org/10.1002/ppap.200900175.
[50] R.P.H. Chang, Plasma oxidation of semiconductor and metal surfaces, in: M. FROMENT (Ed.), Passivity of Metals and Semiconductors, Elsevier, 1983: pp. 437-444. Doi: 10.1016/B978-0-444-42252-1.50070-9.
[51] O. Levasseur, L. Stafford, N. Gherardi, N. Naudé, E. Beche, J. Esvan, P. Blanchet, B. Riedl, A. Sarkissian, Role of substrate outgassing on the formation dynamics of either hydrophilic or hydrophobic wood surfaces in atmospheric-pressure, organosilicon plasmas, Surf. Coat. Technol. 234 (2013) 42–47, https://doi.org/10.1016/j.surfcoat.2013.05.045.
[52] L. Solhi, V. Guccini, K. Heise, I. Solala, E. Niinivaara, W. Xu, K. Mihhels, M. Kröger, Z. Meng, J. Wohlert, H. Tao, E.D. Cranston, E. Kontturi, Understanding nanocellulose-water interactions: turning a detriment into an asset, Chem. Rev. 123 (2023) 1925–2015, https://doi.org/10.1021/acs.chemrev.2c00611.
[53] I.R. Durán, J. Profili, L. Stafford, G. Laroche, Unveiling the origin of the anti-fogging performance of plasma-coated glass: role of the structure and the chemistry of siloxane precursors, Prog. Org. Coat. 141 (2020), https://doi.org/10.1016/j.porgcoat.2019.105401.
[54] J. Wang, D.J. Gardner, N.M. Stark, D.W. Bousfield, M. Tajvidi, Z. Cai, Moisture and oxygen barrier properties of cellulose nanomaterial-based films, ACS Sustain. Chem. Eng. 6 (2018) 49–70, https://doi.org/10.1021/acssuschemeng.7b03523.
[55] F.J.H. Van Assche, S. Unnikrishnan, J.J. Michels, A.M.B. Van Mol, P. Van De Weijer, M.C.M. Van De Sanden, M. Creatore, On the intrinsic moisture permeation rate of remote microwave plasma-deposited silicon nitride layers, Thin Solid Films 558 (2014) 54–61, https://doi.org/10.1016/j.tsf.2014.02.069.